Names, data checks, and new variables

Placeholder names

Name	Stands for
`dat`, `analysis`	data frames
`y`	outcome
`x`, `x1`, `x2`, `x3`, `z`	numerical predictors
`group`	categorical predictor
`m`, `m_multi`, …	fitted models
n, k, G	number of observations, slopes, groups

Read and inspect the data

```
dat <- read.csv("data.csv")

dim(dat)
nrow(dat)
names(dat)
head(dat)
summary(dat)
table(dat$group)
mean(dat$y)
sd(dat$y)
cor(dat$x, dat$y)
tapply(dat$y, dat$group, mean)
```

New variables and the analysis sample

```
dat$x_1k <- dat$x_euros / 1000
dat$group <- factor(dat$group)
dat$d <- as.integer(dat$group == "A")
analysis <- subset(dat, x > 0 & group == "A")
```

Lecture 1: Simple linear regression

Population model and fitted line

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i, \quad \widehat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i.$$

OLS estimates in simple regression

$$\hat{\beta}_1 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2} = \frac{s_{XY}}{s_X^2} = r_{XY}\,\frac{s_Y}{s_X}, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

Residual and sum of squared residuals

$$e_i = Y_i - \widehat{Y}_i, \quad \text{SSE} = \sum_{i=1}^{n} e_i^2.$$

Change of the units of measurement by a factor c

Multiplied by c	New intercept	New slope
outcome only	$c\,\hat{\beta}_0$	$c\,\hat{\beta}_1$
predictor only	$\hat{\beta}_0$	$\hat{\beta}_1 / c$
outcome and predictor	$c\,\hat{\beta}_0$	$\hat{\beta}_1$

Fit a simple regression

```
m <- lm(y ~ x, data = dat)
coef(m)
summary(m)
fitted(m)
residuals(m)
```

Scatterplot with the fitted line

```
plot(y ~ x, data = dat,
     xlab = "Predictor (unit)", ylab = "Outcome (unit)")
abline(m, col = "blue", lwd = 2)
```

Fitted values for chosen values `x_A` and `x_B` of the predictor

```
b <- coef(m)
b[1] + b[2] * x_A

new_cases <- data.frame(x = c(x_A, x_B))
predict(m, newdata = new_cases)
```

Lecture 2: What the model assumes and how residuals reveal it

Residual plot and normal Q-Q plot

```
plot(fitted(m), residuals(m),
     xlab = "Fitted value", ylab = "Residual")
abline(h = 0, lty = 2)

qqnorm(rstandard(m))
qqline(rstandard(m))
```

Standardised residuals and Cook's distance

```
summary(residuals(m))
rstandard(m)
cooks.distance(m)
nobs(m)
```

Flagged observations

$$|\text{standardised residual}| > 3, \quad \text{Cook's distance} > \frac{4}{n}.$$

```
sum(abs(rstandard(m)) > 3)
sum(cooks.distance(m) > 4 / nobs(m))
```

Lecture 3: Coefficient inference, fit, and the F-test

Two-sided test of one coefficient

$$H_0 : \beta_j = 0, \quad H_1 : \beta_j \neq 0, \quad t = \frac{\hat{\beta}_j}{\text{SE}(\hat{\beta}_j)}.$$

Reject H_0 at the 5% level if $|t_{obs}| \geq t_{0.975,\,df}$.

Degrees of freedom and 95% confidence interval, model with k slopes and an intercept

$df = n - k - 1$, $\quad \hat{\beta}_j \pm t_{0.975,\,df}\ \text{SE}(\hat{\beta}_j)$,
simple regression: $k = 1$, $df = n - 2$.

```
summary(m)
confint(m, level = 0.95)
df.residual(m)
qt(0.975, df.residual(m))
```

Labels in the output of `summary(m)`

Label in the output	Symbol
`Estimate`	$\hat{\beta}_j$
`Std. Error`	$\text{SE}(\hat{\beta}_j)$
`t value`	t
`Pr(>|t|)`	two-sided p-value for $H_0 : \beta_j = 0$
`Residual standard error`	s, with $df = n - k - 1$
`Multiple R-squared`	R^2
`Adjusted R-squared`	adjusted R^2 (Lecture 4)
`F-statistic`	overall F, with k and $n - k - 1$ degrees of freedom

The t statistic and the two-sided p-value

```
tab <- summary(m)$coefficients
t_value <- tab["x", "Estimate"] / tab["x", "Std. Error"]
p_value <- 2 * pt(-abs(t_value), df = df.residual(m))
```

Sums of squares and R^2

$$\text{SST} = \sum_i (y_i - \bar{y})^2, \quad \text{SSE} = \sum_i (y_i - \hat{y}_i)^2,$$

$$\text{SSR} = \sum_i (\hat{y}_i - \bar{y})^2, \quad \text{SST} = \text{SSR} + \text{SSE},$$

$$R^2 = 1 - \frac{\text{SSE}}{\text{SST}} = \frac{\text{SSR}}{\text{SST}}.$$

```
y <- dat$y
SST <- sum((y - mean(y))^2)
SSE <- sum(residuals(m)^2)
SSR <- SST - SSE
R2 <- 1 - SSE / SST
summary(m)$r.squared
```

Residual standard error

$$s = \sqrt{\frac{\text{SSE}}{n - k - 1}}.$$

Overall F-test, model with k slopes

$H_0 : \beta_1 = \cdots = \beta_k = 0$,
H_1 : at least one of $\beta_1, \ldots, \beta_k$ is not 0,
$F = \dfrac{\text{SSR}/k}{\text{SSE}/(n-k-1)}$, $df_1 = k,\ df_2 = n-k-1$,
simple regression: $F = t^2$.

```
summary(m)$fstatistic
qf(0.95, df1, df2)
```

Heteroskedasticity-robust standard errors (HC1)

```
library(lmtest)
library(sandwich)

coeftest(m, vcov. = vcovHC(m, type = "HC1"))
coefci(m, vcov. = vcovHC(m, type = "HC1"))
```

Lecture 4: Multiple regression and what controls do

Model with k predictors

$Y_i = \beta_0 + \beta_1 X_{i1} + \cdots + \beta_k X_{ik} + \varepsilon_i$.

```
m_multi <- lm(y ~ x1 + x2 + x3, data = analysis)
summary(m_multi)
confint(m_multi)
cor(analysis[, c("x1", "x2", "x3")])
```

Adjusted R^2, model with k slopes

$$\text{adjusted } R^2 = 1 - (1 - R^2)\,\frac{n-1}{n-k-1}.$$

```
summary(m_multi)$adj.r.squared
```

Fitted difference for changes $\Delta x_1, \ldots, \Delta x_k$ of the predictors

$\Delta\widehat{Y} = \hat\beta_1 \Delta x_1 + \cdots + \hat\beta_k \Delta x_k$.

Fitted difference between two cases A and B

```
cases <- data.frame(
  x1 = c(x1_A, x1_B),
  x2 = c(x2_A, x2_B),
  x3 = c(x3_A, x3_B)
)
fits <- predict(m_multi, newdata = cases)
fits[2] - fits[1]
```

Lecture 5: Categories as predictors and joint tests

Dummy predictor D, reference category $D = 0$

$E[Y_i \mid D_i] = \beta_0 + \beta_1 D_i$, $E[Y_i \mid D_i = 0] = \beta_0$,
$E[Y_i \mid D_i = 1] = \beta_0 + \beta_1$.

Categorical predictor and reference category

```
analysis$group <- factor(analysis$group)
levels(analysis$group)
contrasts(analysis$group)
analysis$group <- relevel(analysis$group,
                          ref = "reference")

m_group <- lm(y ~ x1 + x2 + group, data = analysis)
summary(m_group)
coef(m_group)
confint(m_group)
```

The same model with another reference category

```
analysis$group_alt <- relevel(analysis$group,
                              ref = "other")
m_alt <- lm(y ~ x1 + x2 + group_alt, data = analysis)
all.equal(fitted(m_group), fitted(m_alt))
```

Partial F-test of nested models

Complete model C with k_C slopes. Reduced model R: model C without the q tested terms.

$H_0 : \beta_j = \cdots = \beta_l = 0$ (q coefficients),
H_1 : at least one of $\beta_j, \ldots, \beta_l$ is not 0,
$F = \dfrac{(\text{SSE}_R - \text{SSE}_C)/q}{\text{SSE}_C/(n-k_C-1)}$, $df_1 = q,\ df_2 = n - k_C - 1$.

```
reduced <- lm(y ~ x1 + x2, data = analysis)
complete <- lm(y ~ x1 + x2 + group, data = analysis)
anova(reduced, complete)
```

Labels in the output of `anova(reduced, complete)`

Label in the output	Symbol
`Res.Df`	$n-k-1$ of each model
`RSS`	SSE_R in row 1, SSE_C in row 2
`Df`	q
`Sum of Sq`	$\text{SSE}_R - \text{SSE}_C$
`F`	partial F
`Pr(>F)`	p-value of the partial F-test

One-way ANOVA: one model, G groups

$H_0 : \mu_1 = \mu_2 = \cdots = \mu_G$, $F = \dfrac{\text{SSR}/(G-1)}{\text{SSE}/(n-G)}$,
$df_1 = G-1,\ df_2 = n-G$.

```
m_anova <- lm(y ~ group, data = analysis)
anova(m_anova)
```

Labels in the output of `anova(m_anova)`

Label in the output	Symbol
`Df`	$G-1$ (group), $n-G$ (`Residuals`)
`Sum Sq`	SSR (group), SSE (`Residuals`)
`Mean Sq`	`Sum Sq` / `Df`
`F value`	F
`Pr(>F)`	p-value of the F-test

Lecture 6: Interactions and curvature

Interaction of a numerical predictor X with a dummy D

$\widehat{Y} = \hat\beta_0 + \hat\beta_1 X + \hat\beta_2 D + \hat\beta_3 X D$.

Group	Intercept	Slope
$D = 0$	$\hat\beta_0$	$\hat\beta_1$
$D = 1$	$\hat\beta_0 + \hat\beta_2$	$\hat\beta_1 + \hat\beta_3$

```
m_interaction <- lm(y ~ x * group + z, data = analysis)
summary(m_interaction)
```

Interaction with a numerical moderator Z

$\widehat{Y} = \hat\beta_0 + \hat\beta_1 X + \hat\beta_2 Z + \hat\beta_3 X Z$.
Slope of X at $Z = z$: $\hat\beta_1 + \hat\beta_3 z$.

```
m_moderator <- lm(y ~ x * z, data = analysis)
summary(m_moderator)
```

Centred predictor

```
analysis$x_c <- analysis$x - mean(analysis$x)
m_centred <- lm(y ~ x_c * group + z, data = analysis)
```

Quadratic model

$\widehat{Y} = \hat\beta_0 + \hat\beta_1 X + \hat\beta_2 X^2$.
Fitted change from x to $x+1$: $\hat\beta_1 + \hat\beta_2(2x+1)$.

```
m_linear <- lm(y ~ x + z, data = analysis)
m_quadratic <- lm(y ~ x + I(x^2) + z, data = analysis)
summary(m_quadratic)
anova(m_linear, m_quadratic)
```

Commands by purpose

Purpose	R command
Fit a model	`lm(y ~ predictors, data = dat)`
Coefficient table and fit	`summary(m)`
Coefficients only	`coef(m)`
95% confidence intervals	`confint(m)`
Fitted values and residuals	`fitted(m)`, `residuals(m)`
Summary of the residuals	`summary(residuals(m))`
Standardised residuals	`rstandard(m)`
Cook's distance	`cooks.distance(m)`
Fitted values for new cases	`predict(m, newdata = ...)`
R^2, adjusted R^2, overall F	`summary(m)`
Residual degrees of freedom	`df.residual(m)`
Critical value of a 95% interval	`qt(0.975, df.residual(m))`
Critical value of a 5% F-test	`qf(0.95, df1, df2)`
Partial F-test of nested models	`anova(reduced, complete)`
ANOVA table of one model	`anova(m)`
Reference category of a factor	`relevel(factor(group), ref = "category")`
Interaction	`x * group`
Square in a formula	`I(x^2)`
Robust standard errors	`coeftest(m, vcov. = vcovHC(m, type = "HC1"))`